

A CRITICAL ANALYSIS OF SPATIAL ANALYSIS

INTRODUCTION - SPATIAL ANALYSIS

Spatial analysis is a set of techniques and methods used to study the spatial patterns and relationships that exist within data. It involves the use of tools and methods from a variety of fields, including geography, statistics, and computer science, to analyze and interpret spatial data. The goal of spatial analysis is to provide a deeper understanding of the spatial patterns and relationships that exist within data, and to use this understanding to support decision-making and problem solving in a variety of fields. Some common applications of spatial analysis include environmental planning, transportation planning, and crime mapping.(Oxford Dictionary, 2017)

DEFINITIONS OF SPATIAL ANALYSIS

According to ESRI's (2013) definition, this type of analysis is essential to the process of problem-solving and helpful when attempting to do so. Find patterns and links in the data, as well as how they might be used for a variety of purposes, such as analysing the distribution of crime in a city to find out where to increase the presence of police officers. Another example of this would be determining how the data might be used would be to find out how they could be used for a variety of purposes.

In their statement from 1997, Xiang and Han provided further clarification on this idea. It is based on the formative definition and classification of objects in geographical space, and it depicts the spatial relationship and action of objectives as well as an analysis of their spatial interaction. Additionally, it analyses how objectives interact with one another spatially. The topic of geographical analysis includes a subsection known as spatial analysis. In addition to this, it is a practical system that provides support with the process of making decisions pertaining to the use of space. (Xiang & Han, 1997)

This description goes further than the one provided by ESRI and provides a more accurate summary of what is involved in spatial analysis than the original definition. In addition to this, it underlines the significance of a geographical place and how the many entities that exist within it interact with one another. In the last part of this section, I will elaborate on how spatial analysis can be used to investigate the interactions between objects and activities that take place within a particular location, and will emphasize how spatial analysis enables the development of a variety of different types of systems.

Both Goodchild and Longley offer two distinct explanations of what they mean when they use the phrase "spatial analysis." In spite of the fact that it is a broad concept, it brings to light the potential significance of geographical data. In the case that the geographical situation changes, it will be necessary to determine which type of statistical analysis is the most appropriate to use. After that, they offer a definition that is more all-encompassing.

The impact that alterations to spatial data will have on analytical processes and the findings obtained from such processes is the primary subject of this discussion. These concepts highlight the need for spatial data and how this data can be used to establish a grasp of a problem or question in a specific geographical location by focusing on how the data may be applied to the problem or question. The spatial data can be used to establish a grasp of the problem or question by focusing on how the data may be applied to the problem or question.

In 2003, a study authored by Haining and Haining and published in that same year revealed the idea of spatial analysis that they had developed. In the process of spatial analysis, data are utilised that provide detailed information regarding the spatial interactions that occur between various items in a particular location on Earth. This observation is in line with the subject matter, which states that spatial analysis needs to make use of spatial data that is geographically referenced in a particular space and that there are a variety of methodologies and methods for analysing such data. This observation is in line with the subject matter because it is in line with the statement that spatial analysis needs to make use of spatial data that is geographically referenced in a particular space.

EXAMPLES OF SPATIAL ANALYSIS

There are many different techniques that can be used in spatial analysis. Some examples of spatial analysis techniques include:

Location-based data analysis: the analysis of data with a spatial or geographical component, such as sales data from retail stores or population data from census data.

Spatial clustering and segmentation: the process of identifying clusters or groups of data points that are geographically close to each other, or the process of dividing a geographic area into homogeneous segments based on certain criteria.

Spatial regression analysis: the use of regression analysis to model relationships between dependent and independent variables, taking into account the spatial location of the data points.

Spatial interpolation and extrapolation: the process of estimating values at locations where data is not available, based on data from surrounding locations.

Spatial autocorrelation analysis: the analysis of the spatial pattern and relationships of a set of data points, to determine whether the values of a variable at a given location are influenced by the values at neighboring locations.

Network analysis and routing: the analysis of interconnected systems, such as transportation networks or utility networks, to optimize routes, minimize costs, and improve efficiency.

Visualization of geographical data: the representation of geographical data in a visual form, such as maps, charts, or graphs, to facilitate understanding and analysis.

Terrain analysis and modeling: the analysis of topographic data to determine surface characteristics and terrain features, such as slope and aspect, for applications in civil engineering, agriculture, and natural resource management.

Spatial database management: the organization, storage, and retrieval of spatial data in a database, to support efficient data analysis and visualization.

COMPARISONS BETWEEN SPATIAL AND NON SPATIAL ANALYSIS

Spatial and non-spatial analysis are two different types of analysis that are used to study data. Spatial analysis is focused on the spatial patterns and relationships that exist within data, while non-spatial analysis is focused on the non-spatial, or non-spatial, aspects of data. Some key differences between these two types of analysis include:

Focus: Spatial analysis focuses on the spatial patterns and relationships within data, while non-spatial analysis focuses on the non-spatial aspects of data.

Tools and methods: Spatial analysis uses a range of tools and methods from fields such as geography, statistics, and computer science, while non-spatial analysis may use tools and methods from fields such as statistics, computer science, and data mining.

Data types: Spatial analysis is typically used with data that has a spatial component, such as geographic coordinates or addresses, while non-spatial analysis can be used with any type of data, regardless of whether it has a spatial component.

Applications: Spatial analysis is commonly used in fields such as environmental planning, transportation planning, and crime mapping, while non-spatial analysis is commonly used in fields such as marketing, finance, and healthcare.

Overall, spatial and non-spatial analysis are two different approaches to data analysis that are used to study different aspects of data.

SPATIAL ANALYSIS TECHNIQUES

1. Proximity Analysis: Buffer Analysis

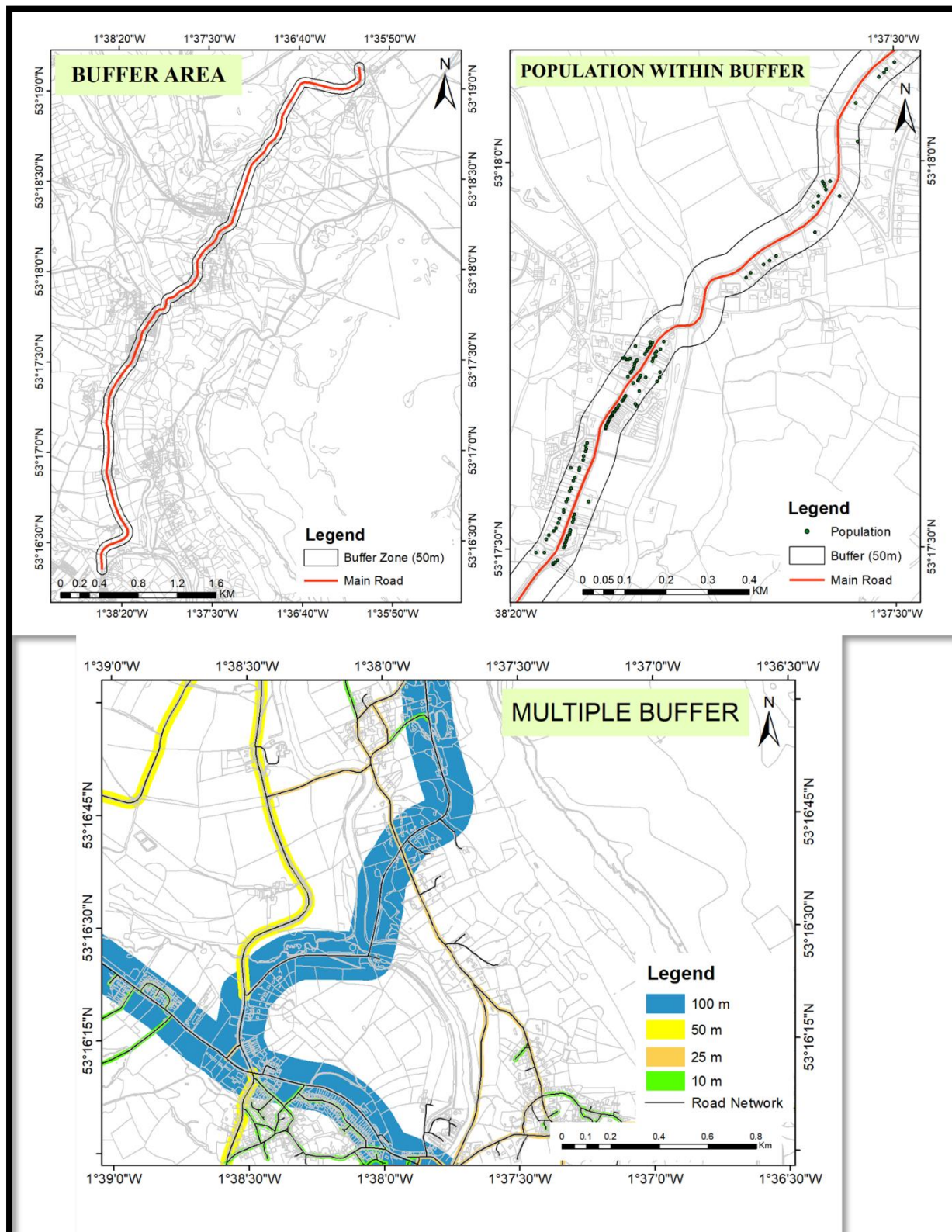
Proximity analysis is a spatial technique that measures the relationship between two or more spatial features based on their proximity to each other. A buffer analysis is a type of proximity analysis that creates a buffer zone around one or more features. The buffer zone is a zone of a certain distance around a feature, and is used to identify other features that are within a certain proximity of that feature. Buffer analysis is commonly used for site selection, network analysis, and environmental impact assessments.

Buffer analysis in ArcGIS is a spatial analysis technique used to create buffer zones around features in a map. The buffer zones can be used to measure distances, analyze relationships between features, and identify areas of influence. Here's how to perform buffer analysis in ArcGIS:

- Load the feature layer you want to analyze into ArcMap.
- Select the feature layer in the Table of Contents and right-click to open the context menu.
- Select Analysis > Proximity > Buffer.
- In the Buffer wizard, select the feature layer you want to analyze and specify the buffer distance.
- Choose a unit of measurement (e.g. feet, meters, kilometers) and select the type of buffer you want to create (e.g. round, square, donut).
- Optionally, you can specify the number of segments used to approximate curves in the buffer.
- Choose an output location for the buffer layer and click OK.

The buffer layer will be created and added to the map as a new layer. You can style the layer and perform additional analyses on the buffer layer.

Note: Buffer analysis is just one of many spatial analysis tools available in ArcGIS. The software has a variety of other tools for performing complex spatial analyses, including overlays, intersections, and proximity analysis.



2. Mean Centre and Standard Distance

Measuring distribution refers to the quantification of the spatial arrangement and dispersion of features on a map.

Mean center, also known as centroid, is a common method for measuring the center of a distribution. It's calculated by finding the average of all x and y coordinates of features, producing a single point that represents the center of the distribution.

Standard distance, also known as standard deviation, measures the degree of dispersion or spread of features from the mean center. It's calculated by finding the average distance of features from the mean center, with larger values indicating a more spread out distribution.

Both mean center and standard distance are commonly used to assess spatial patterns, identify clusters and outliers, and inform decision making in fields such as urban planning and resource management.

Mean Center and Standard Distance are spatial analysis tools in ArcGIS that are used to identify the central location and spread of a set of features on a map. Here's how to perform Mean Center and Standard Distance analysis in ArcGIS:

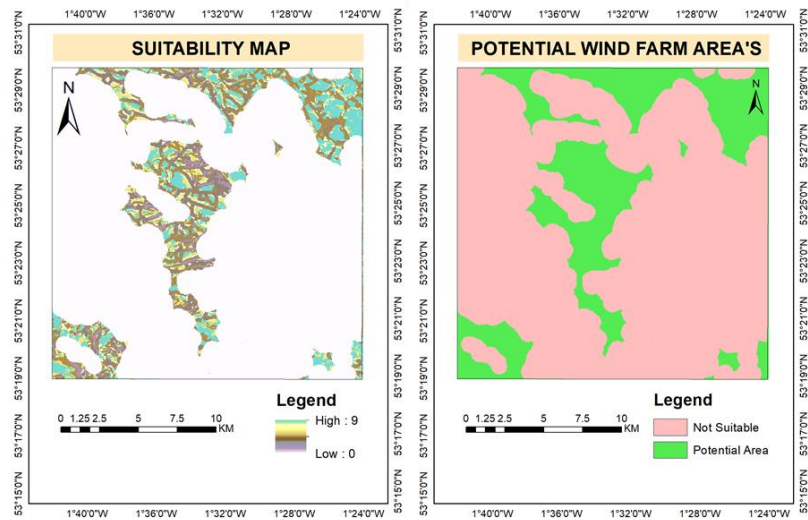
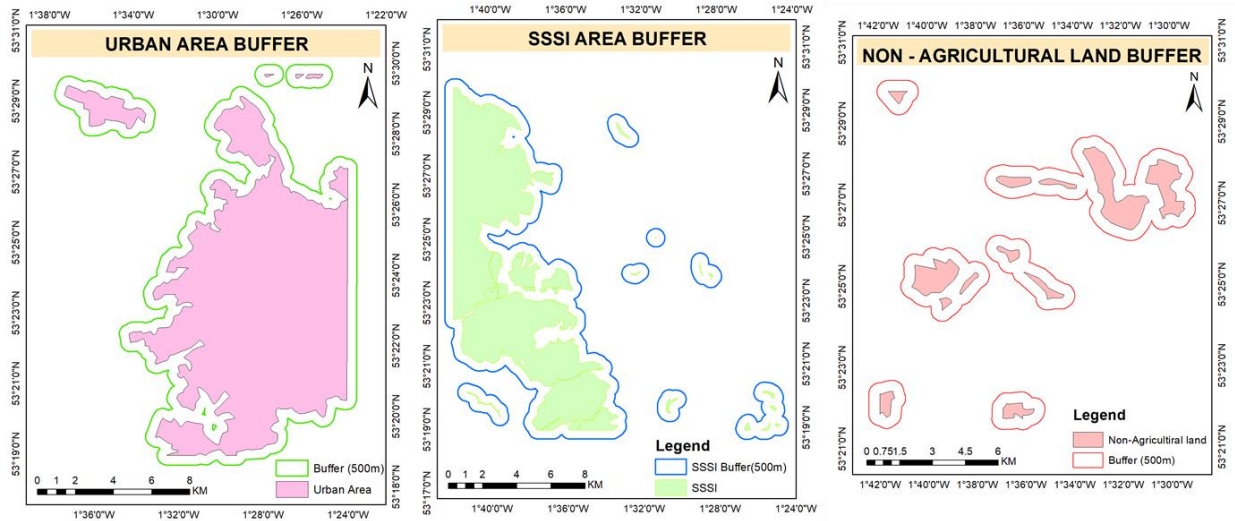
Mean Center:

- Load the feature layer you want to analyze into ArcMap.
- Select the feature layer in the Table of Contents and right-click to open the context menu.
- Select Analysis > Statistics > Mean Center.
- In the Mean Center wizard, select the feature layer you want to analyze and specify an output location for the result.
- Choose a field from the layer to weight the analysis (optional).
- Click OK to run the analysis.
- The Mean Center result will be added to the map as a new layer. The result shows the central location of the features, represented as a point.

Standard Distance:

- Load the feature layer you want to analyze into ArcMap.
- Select the feature layer in the Table of Contents and right-click to open the context menu.
- Select Analysis > Statistics > Standard Distance.
- In the Standard Distance wizard, select the feature layer you want to analyze and specify an output location for the result.
- Choose a field from the layer to weight the analysis (optional).
- Click OK to run the analysis.
- The Standard Distance result will be added to the map as a new layer. The result shows the spread of the features, represented as an ellipse centered on the mean center.

Note: Mean Center and Standard Distance are just two of many spatial analysis tools available in ArcGIS. The software has a variety of other tools for performing complex spatial analyses,



REFERENCE

- Heywood, D., Cornelius, S., & Carver, S. (2011). An introduction to geographical information systems (4th ed.). Prentice Hall.
- Smith (2015). Geospatial Analysis: A Comprehensive Guide to Principles Techniques and Software Tools. Winchelsea Press: Winchelsea.
- Oxford Dictionary. (2017). Analysis. <https://en.oxforddictionaries.com/>
- ESRI. (2013). The Language of Spatial Analysis.
- Xiang, N., & Han, X. (1997). Definition and contents of spatial analysis. Journal of Central South University of Technology, 4(1), 28-31.
- Goodchild, M. F., & Longley, P. A. (1999). The future of GIS and spatial analysis. Geographical information systems, 1, 567-580.
- Haining, R. P., & Haining, R. (2003). Spatial data analysis: theory and practice. Cambridge university press.
- Dawson, Q., Kechavarzi, C., Leeds-Harrison, P. B., & Burton, R. G. O. (2010). Subsidence and degradation of agricultural peatlands in the Fenlands of Norfolk, UK. Geoderma, 154(3-4), 181-187.
- De Smith, M. J., Goodchild, M. F., & Longley, P. (2007). Geospatial analysis: a comprehensive guide to principles, techniques and software tools. Troubador publishing limited
- Fotheringham, A. S., & Rogerson, P. A. (1993). GIS and spatial analytical problems. International Journal of Geographical Information Science, 7(1), 3-19
- Longley et al, P. (2005). Geographical information systems and science. London: Wiley.
- Mitchell, A. (1999). The ESRI Guide to GIS: Volume 1 Geographic Patterns & Relationships ESRI New York. Redlands, California: Environmental Systems Research Institute.
- Howell, F. M., Porter, J. R., & Matthews, S. A. (2016). Recapturing Space: New Middle-Range Theory in Spatial Demography. Cham Springer International Publishing.
- John, M., Goodchild, M. F., & Longley, P. (2007). Geospatial analysis : a comprehensive guide to principles, techniques and software tools. Leicester Matador Cop.
- Longley, P., Goodchild, M. F., Maguire, D. J., & Rhind, D. W. (2001). Geographic Information Systems and Science. John Wiley & Sons Incorporated.
- Ripley, B. D. (2005). Spatial Statistics. John Wiley & Sons.
- Anselin, L. (1995). Local Indicators of Spatial Association - LISA. Geographical Analysis, 27(2), 93-115. <https://doi.org/10.1111/j.1538-4632.1995.tb00338.x>
- Datta, A., & Datta, D. (2019). Spatial Statistics and ArcGIS Pro: An Introduction to Applied Spatial Data Analysis. Redlands, CA: Esri Press.
- Ripley, B. D. (1977). Modelling spatial patterns. Journal of the Royal Statistical Society. Series B (Methodological), 39(2), 172-212. <https://doi.org/10.1111/j.2517-6161.1977.tb01236.x>

- Sun, J., & Cui, J. (2010). An improved method for spatial cluster analysis using K-function. *Journal of Geographical Sciences*, 20(3), 463-471. <https://doi.org/10.1007/s11442-010-0463-7>

